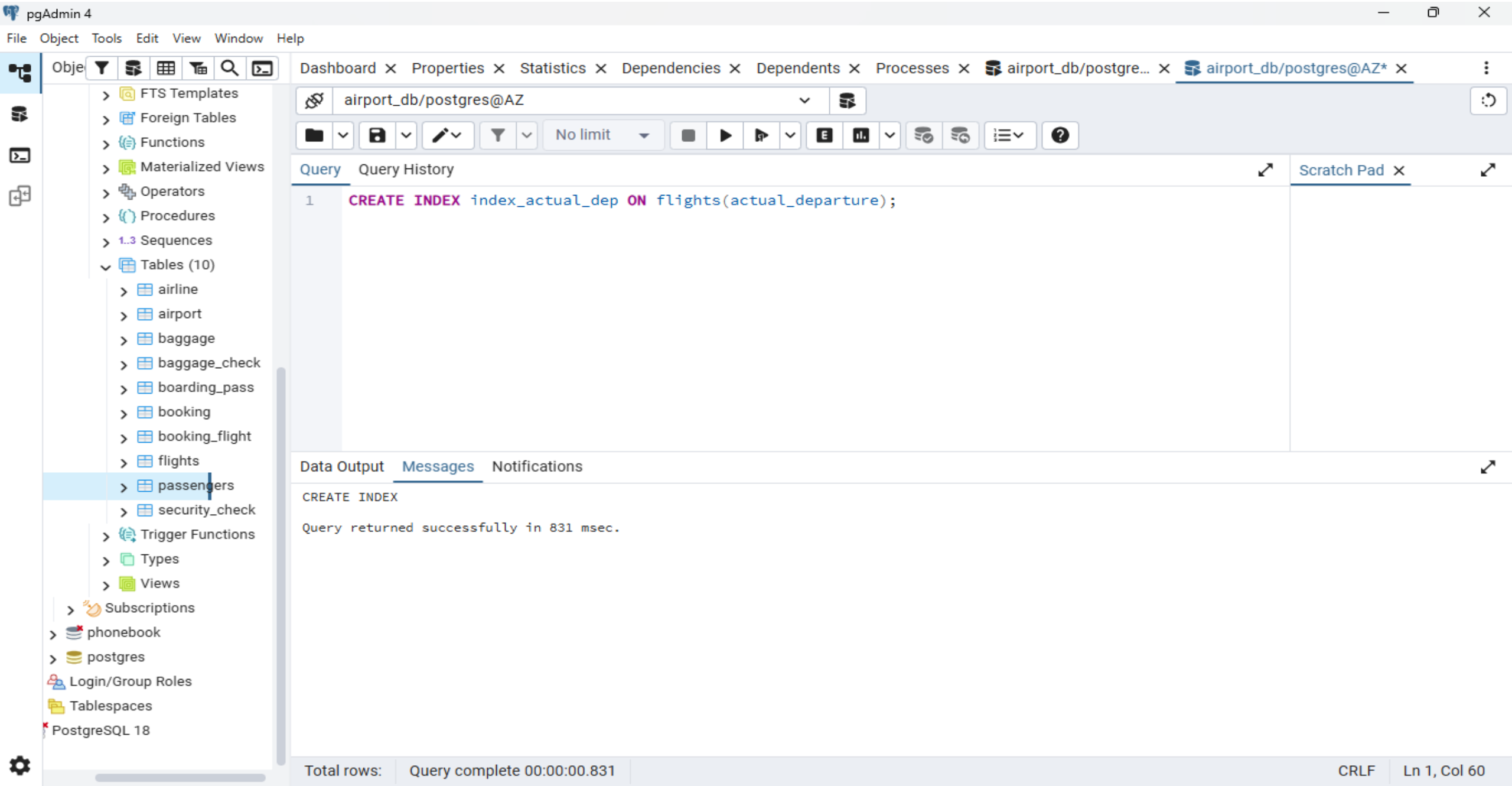
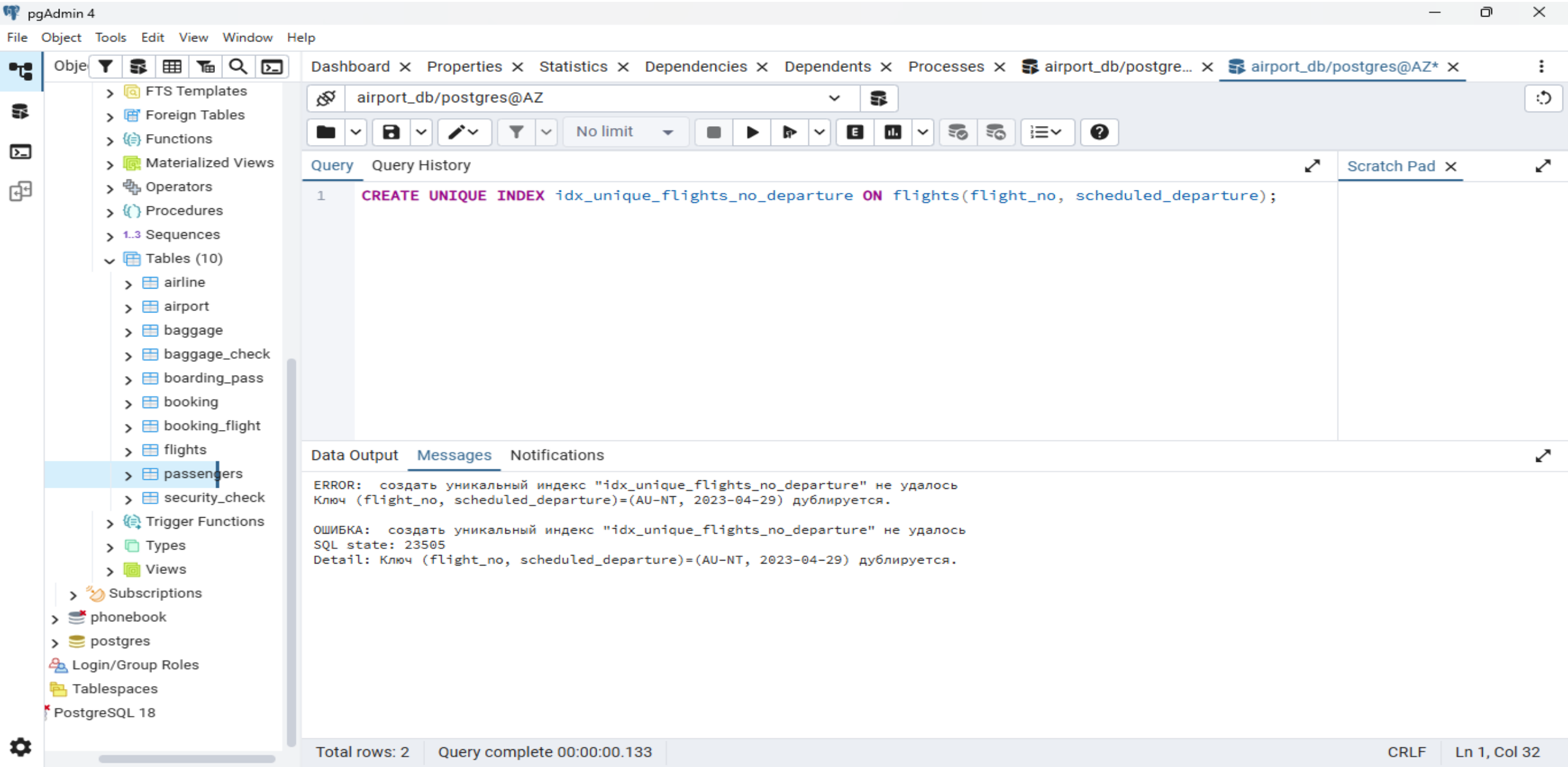
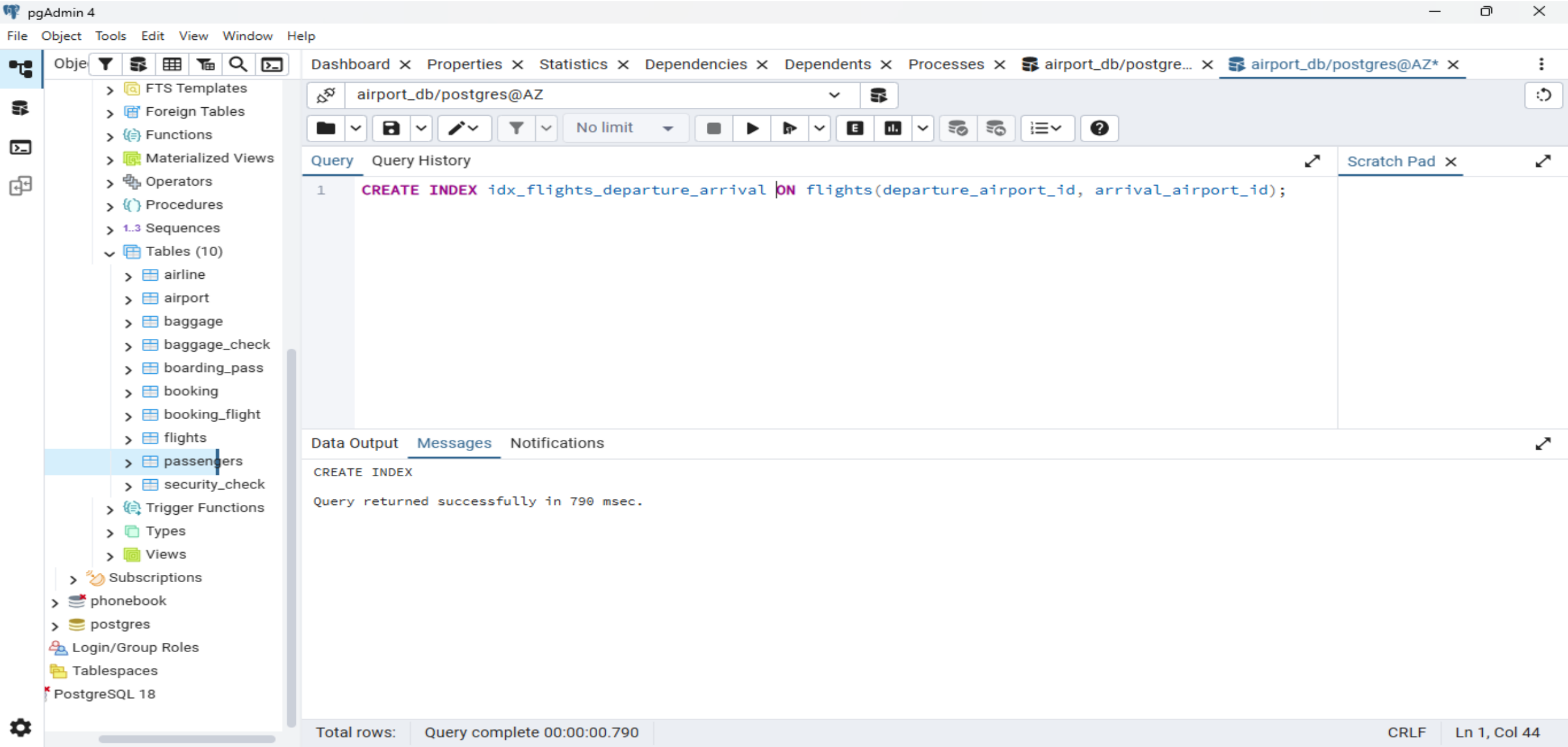
1.Create an index on the actual_departure column in the flights table.



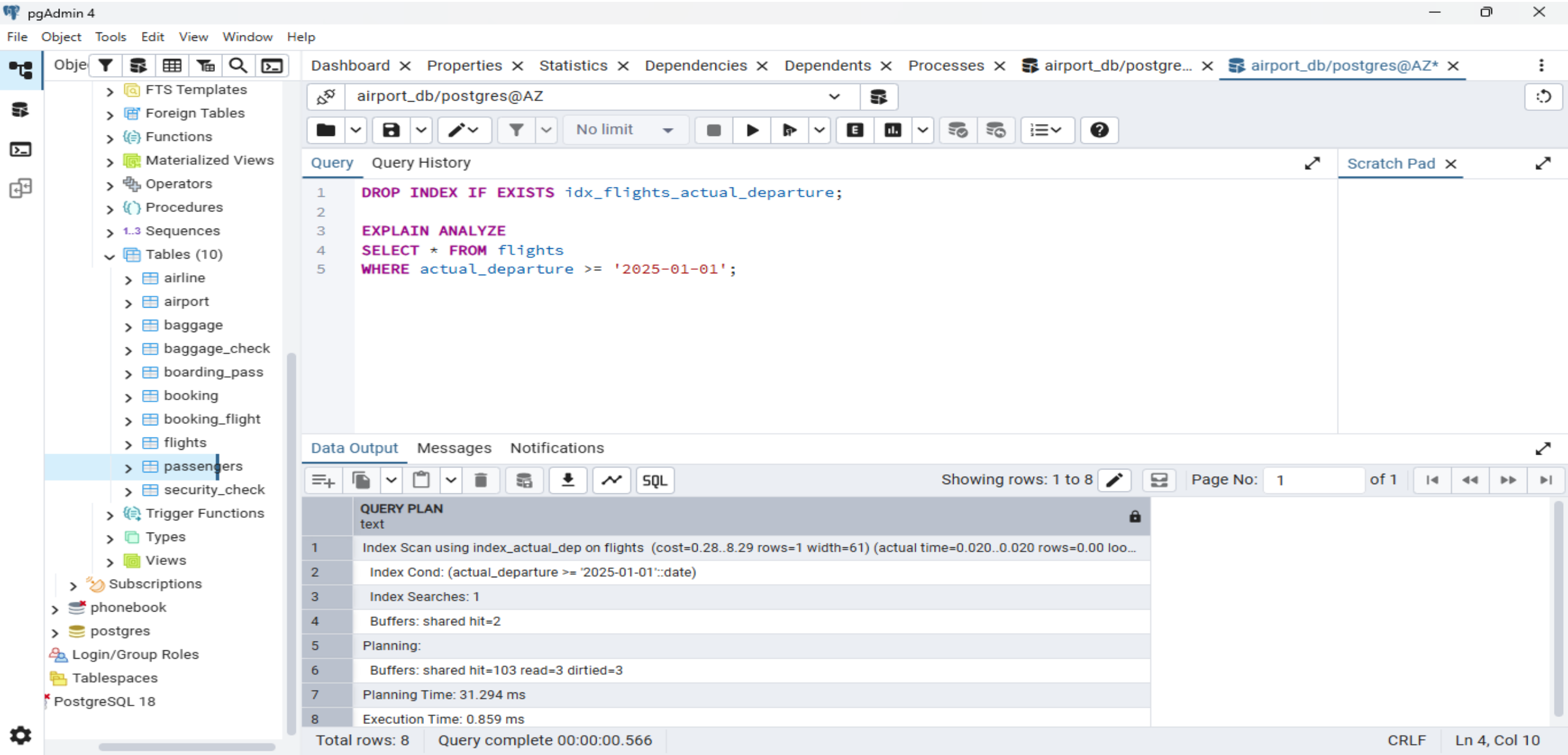
2.Create a unique index to ensure flight_no and scheduled_departure combinations are unique.



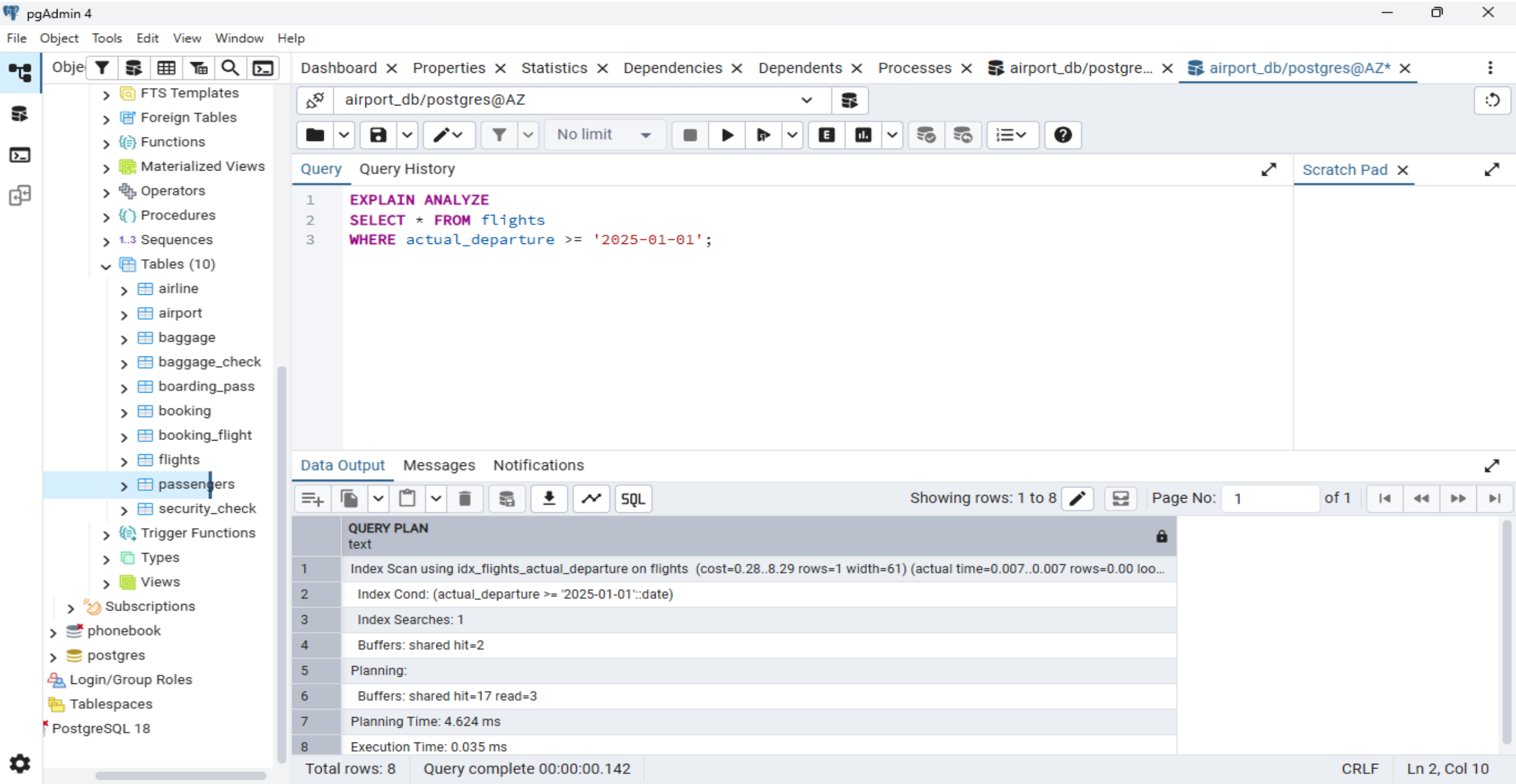
3. Create a composite index on the departure_airport_id and arrival_airport_id columns.



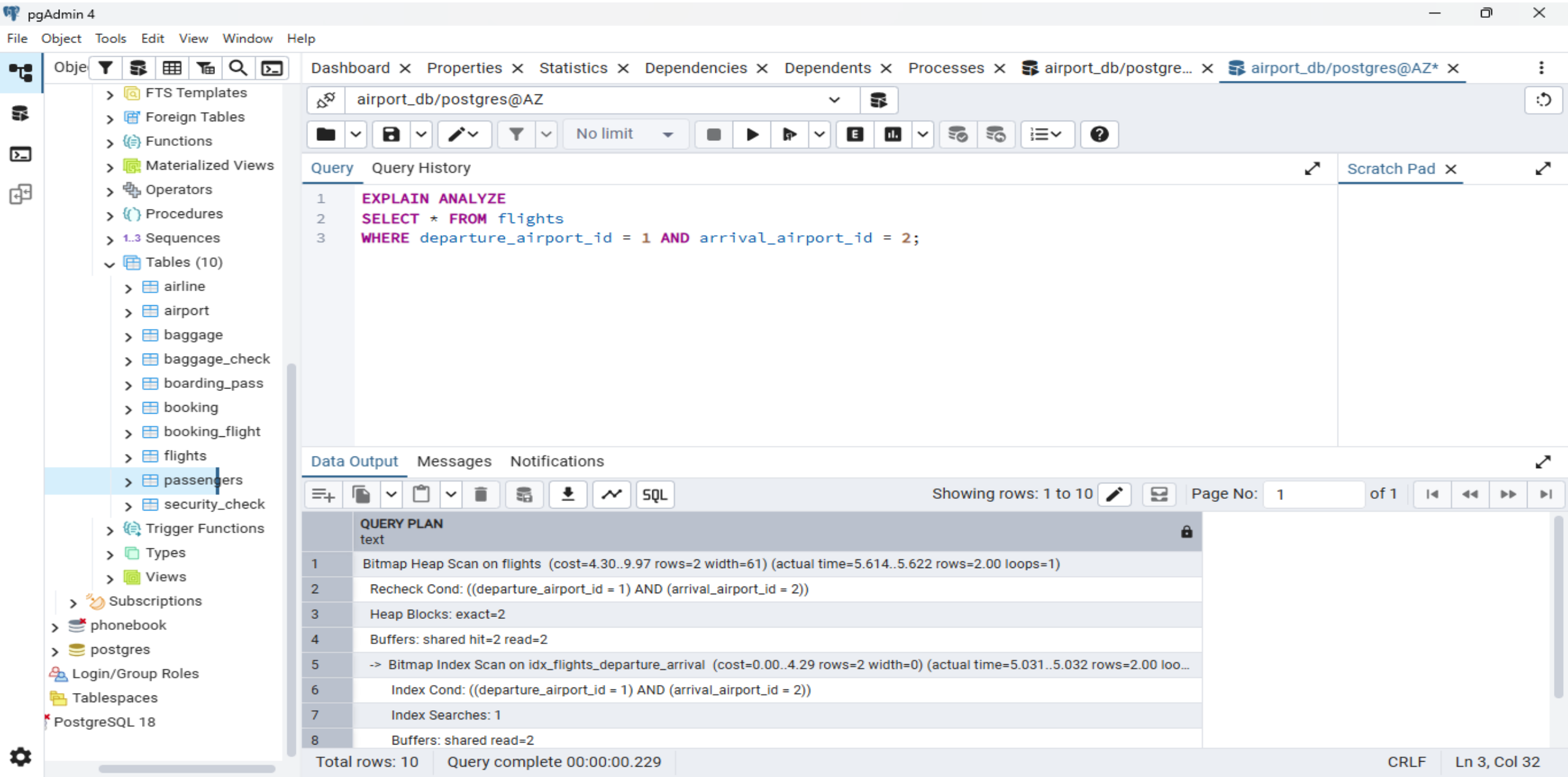
4. Evaluate the difference in query performance with and without indexes. Measure performance differences.



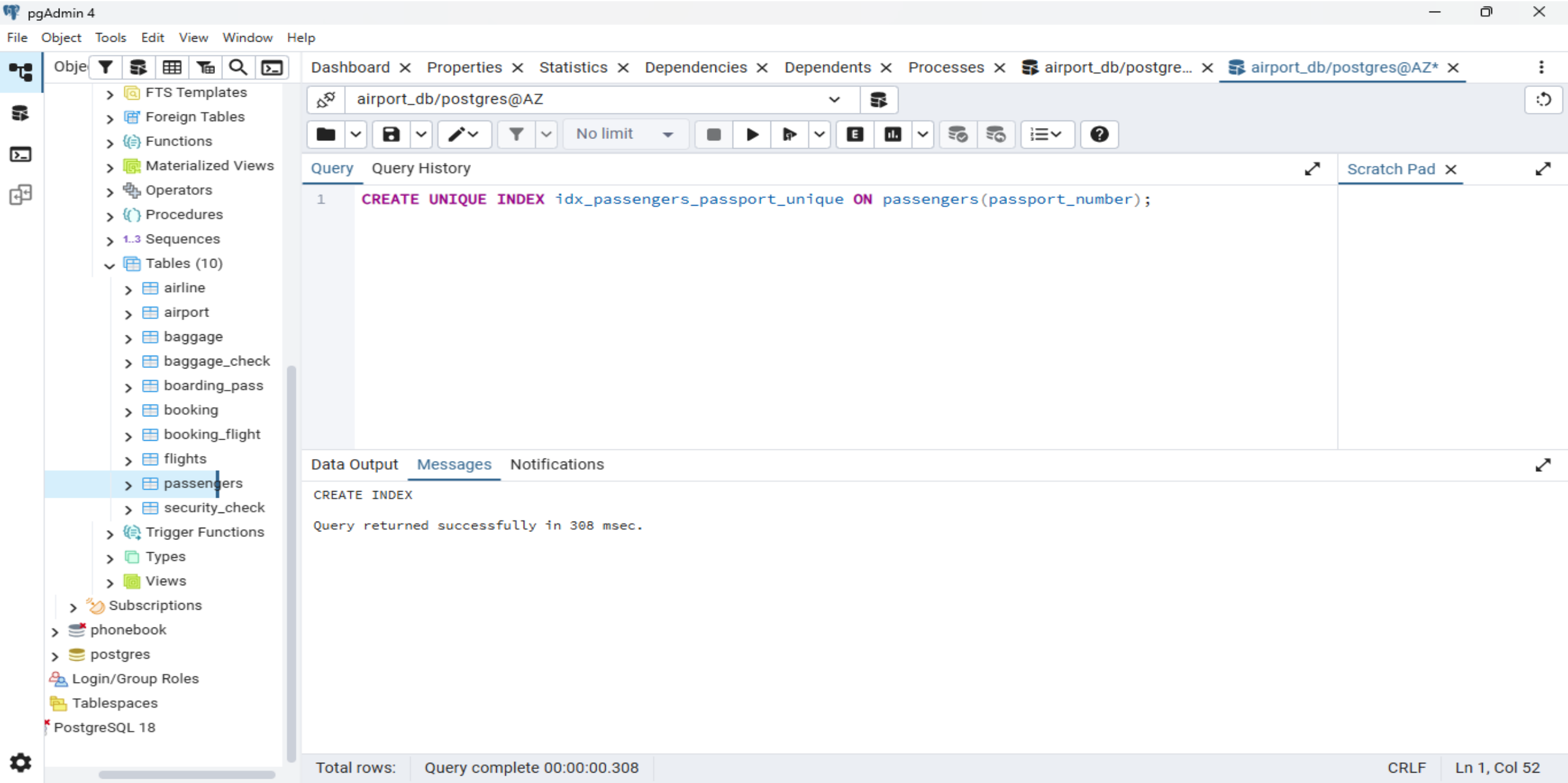
With the index:



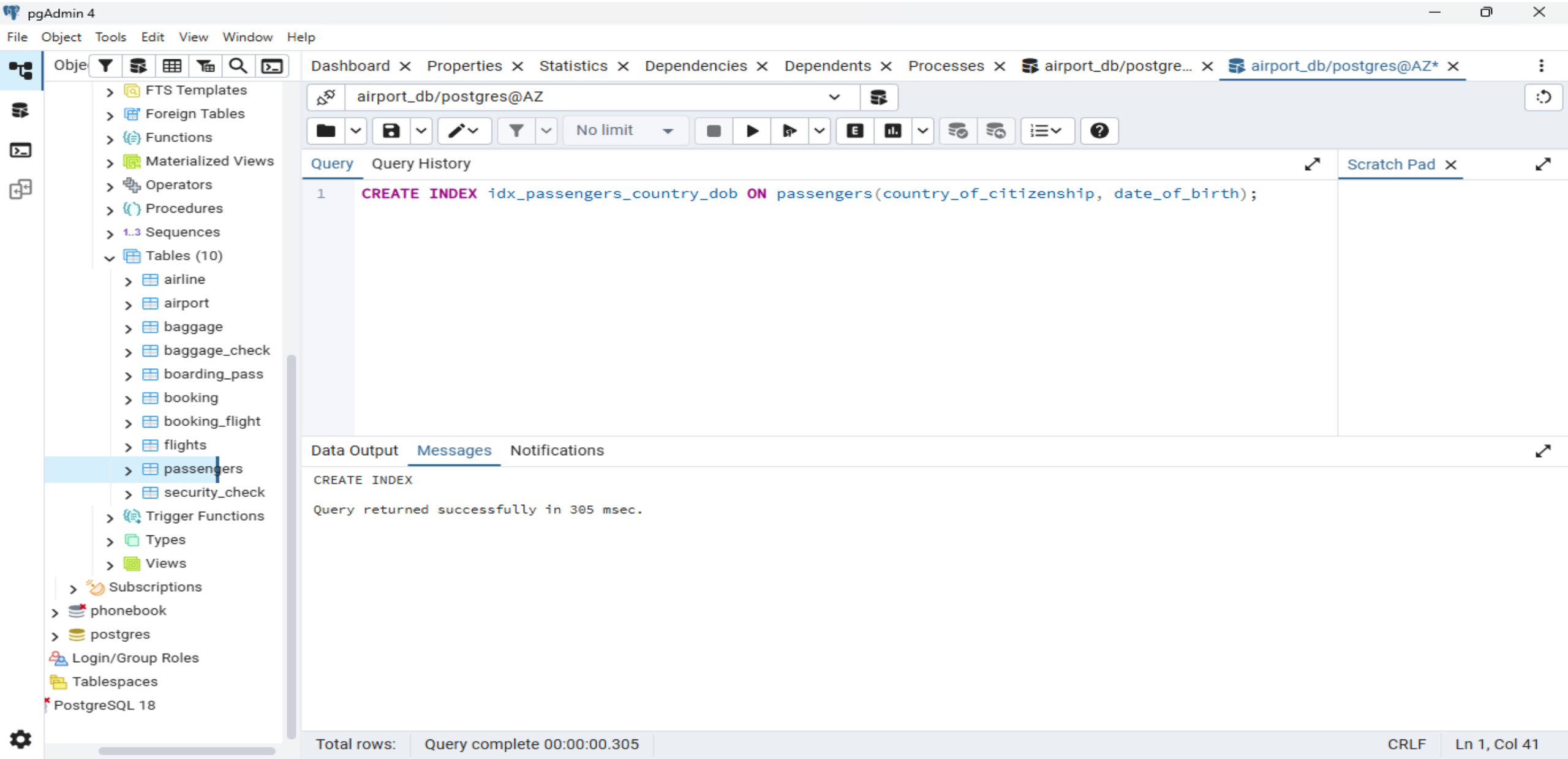
5. Use EXPLAIN ANALYZE to check index usage in a query filtering by departure_airport and arrival_airport.



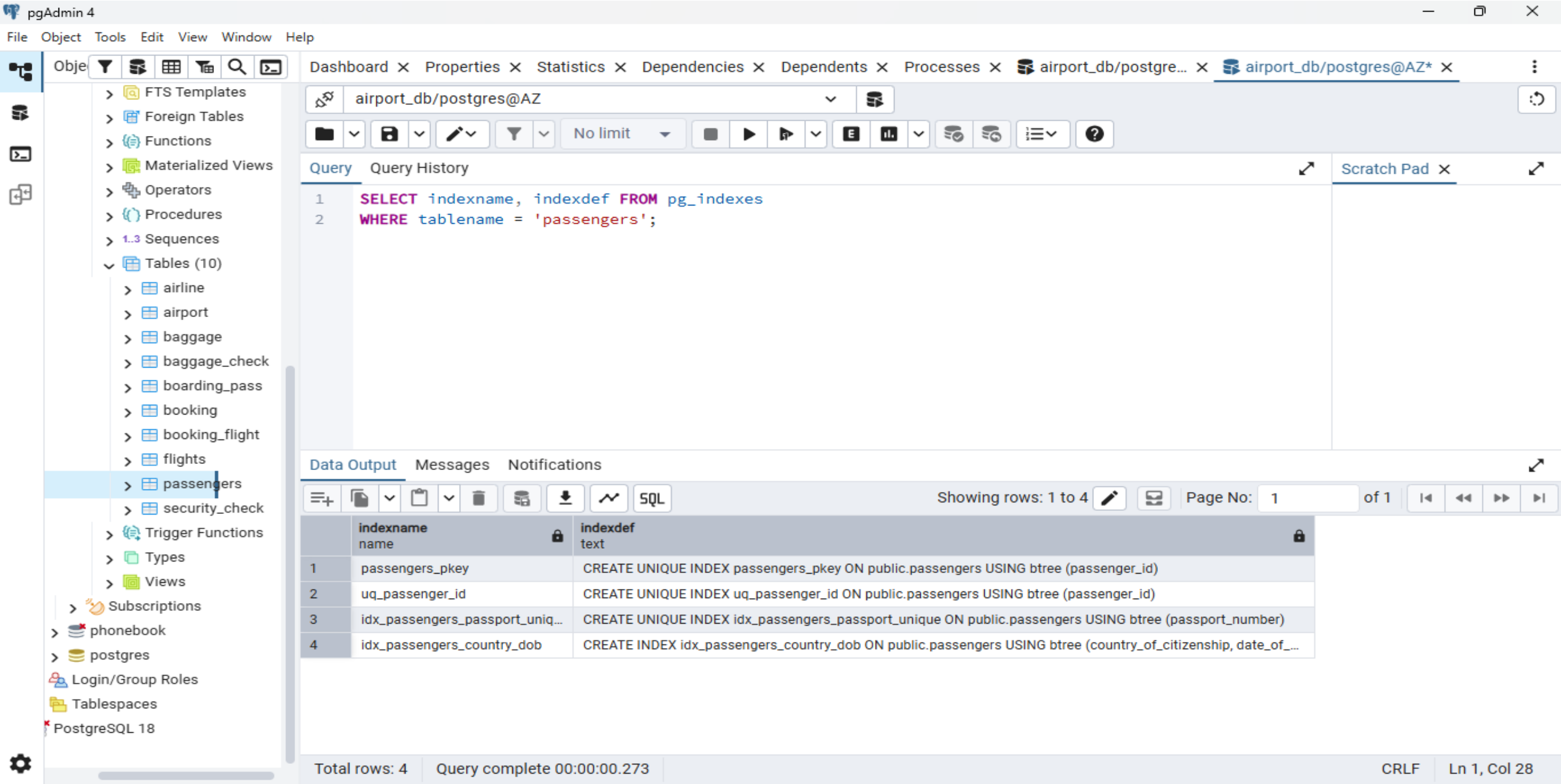
6.Create a unique index for the passport_number of the Passengers table.



7. Create an index for the Passengers table. Use for that first name, last name, date of birth and country of citizenship. Then, write a SQL query to find a passenger who was born in Philippines and was born in 1984 and check if the query uses indexes or not. Give an explanation of the results.



8. Write a SQL query to list indexes for table Passengers. After, delete the created indexes.



pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  DROP INDEX IF EXISTS idx_passengers_passport_unique;
2  DROP INDEX IF EXISTS idx_passengers_country_dob;
```

Scratch Pad

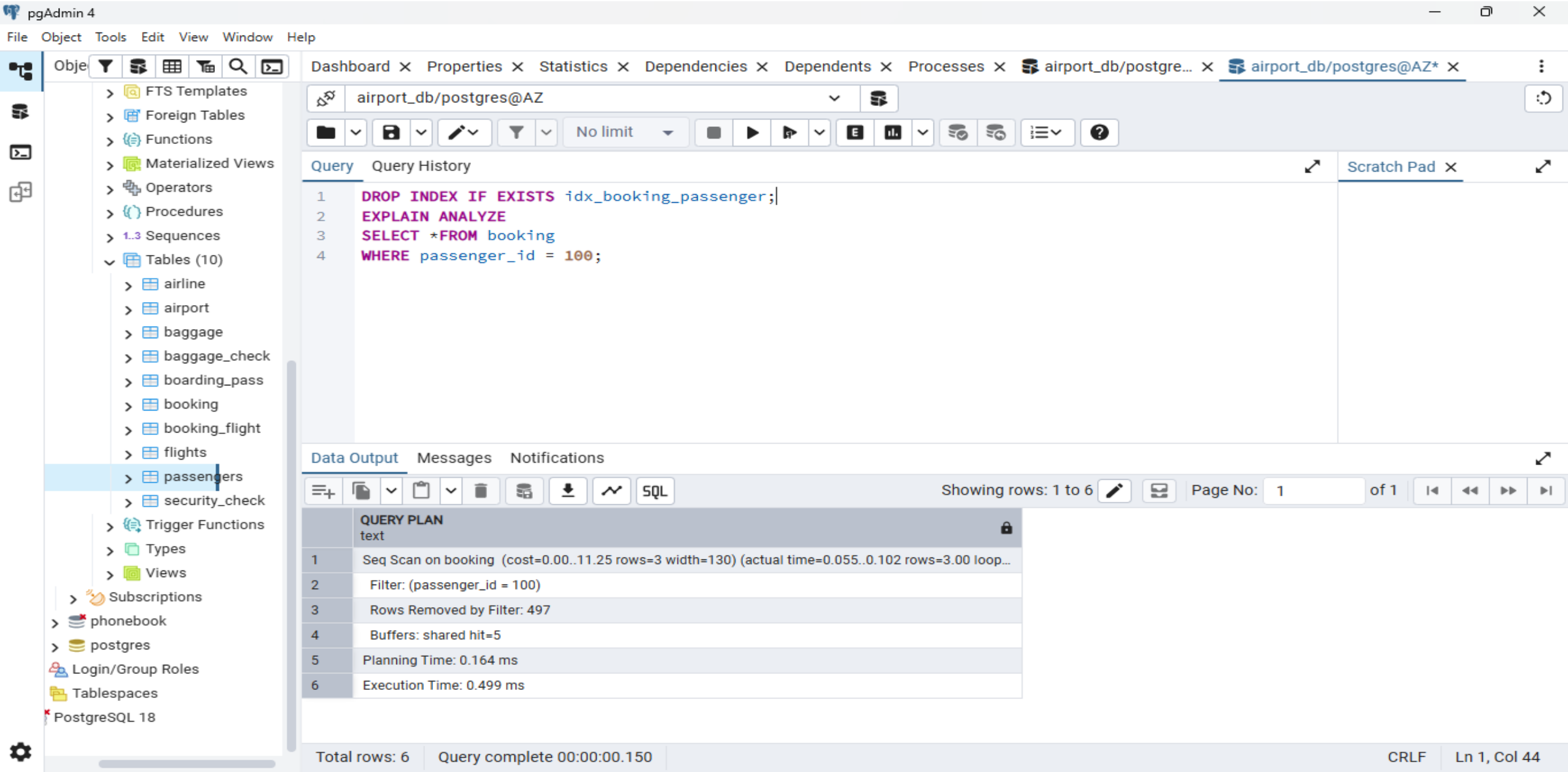
Data Output | Messages | Notifications

```
DROP INDEX

Query returned successfully in 261 msec.
```

Total rows: | Query complete 00:00:00.261 | CRLF | Ln 1, Col 53

9. Create an index on the passenger_id column of the booking table. Using EXPLAIN ANALYZE, compare the execution plan and execution time of a query that selects all bookings of one passenger (for example, passenger_id = 100) before and after creating the index



pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

Scratch Pad

```
1  EXPLAIN ANALYZE
2  SELECT * FROM booking
3  WHERE passenger_id = 100;
```

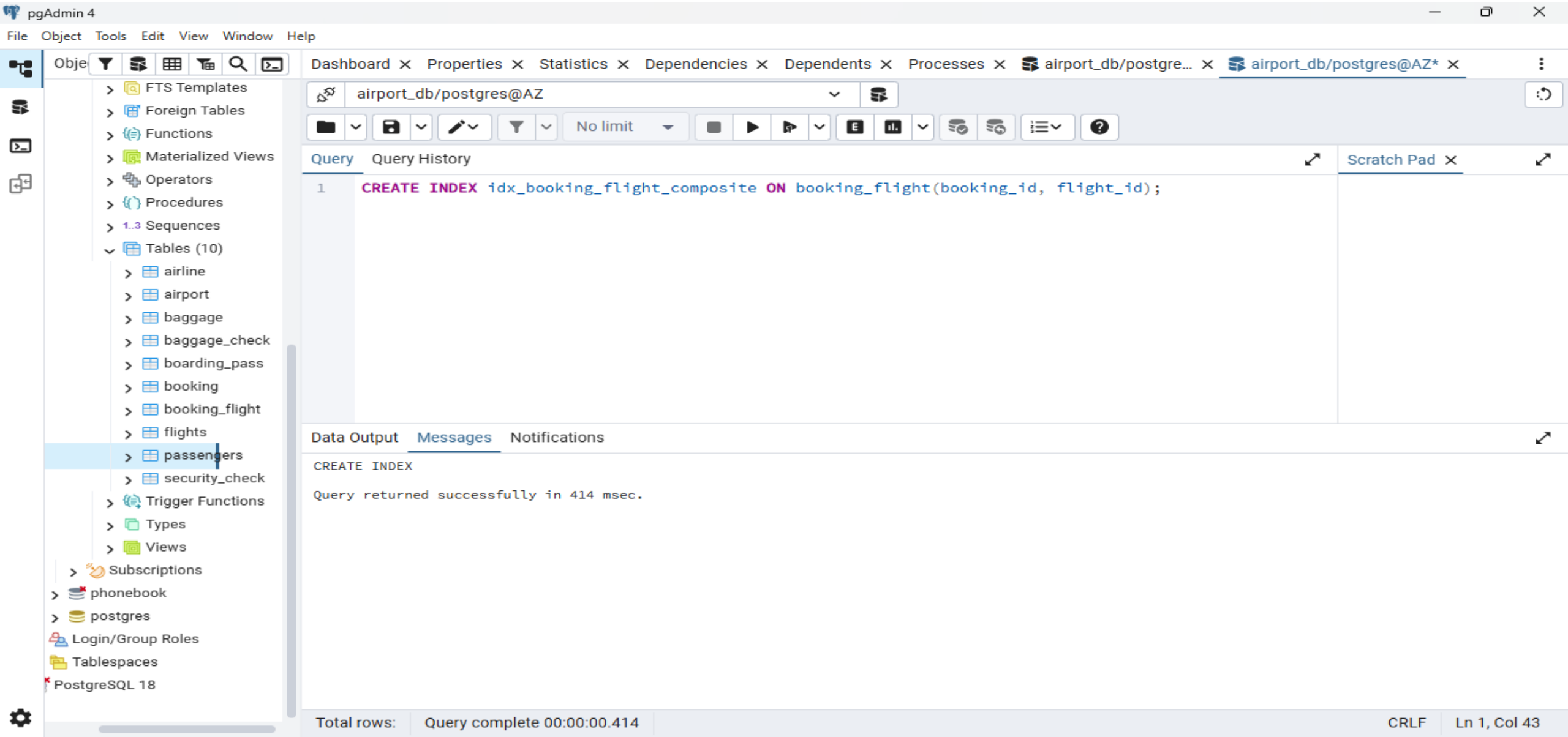
Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on booking (cost=4.17..9.24 rows=3 width=130) (actual time=0.857..0.867 rows=3.00 loops=1)
2	Recheck Cond: (passenger_id = 100)
3	Heap Blocks: exact=2
4	Buffers: shared hit=2 read=1
5	-> Bitmap Index Scan on idx_booking_passenger (cost=0.00..4.17 rows=3 width=0) (actual time=0.756..0.756 rows=3.00 loo...
6	Index Cond: (passenger_id = 100)
7	Index Searches: 1
8	Buffers: shared read=1

Total rows: 12 | Query complete 00:00:00.205 | CRLF | Ln 2, Col 10

10. Create a composite index on the booking_flight table using the booking_id and flight_id columns. Write a query that joins booking, booking_flight and flights and returns the flight_no and price for bookings with a price greater than 9000. Check with EXPLAIN ANALYZE which indexes are used in each step of the join.



pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT f.flight_no, b.price FROM booking b
3  JOIN booking_flight bf ON b.booking_id = bf.booking_id
4  JOIN flights f ON bf.flight_id = f.flight_id
5  WHERE b.price > 9000;
```

Scratch Pad

Data Output | Messages | Notifications

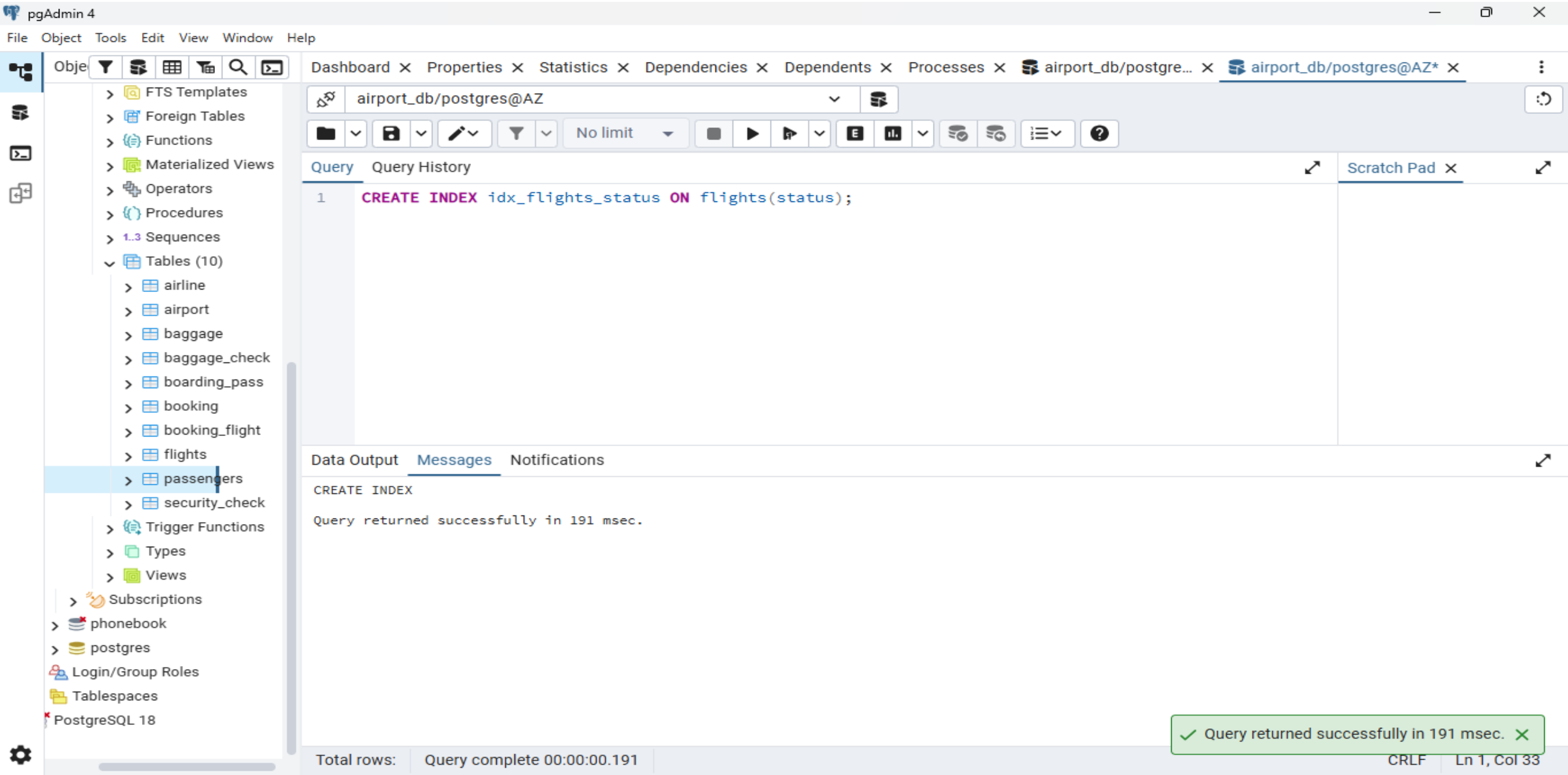
Showing rows: 1 to 24 | Page No: 1 of 1

	QUERY PLAN text
1	Hash Join (cost=33.15..58.37 rows=120 width=12) (actual time=1.366..2.972 rows=118.00 loops=1)
2	Hash Cond: (f.flight_id = bf.flight_id)
3	Buffers: shared hit=25
4	-> Seq Scan on flights f (cost=0.00..21.02 rows=802 width=10) (actual time=0.037..1.507 rows=802.00 loops=1)
5	Buffers: shared hit=13
6	-> Hash (cost=31.65..31.65 rows=120 width=10) (actual time=1.252..1.256 rows=118.00 loops=1)
7	Buckets: 1024 Batches: 1 Memory Usage: 14kB
8	Buffers: shared hit=12

Total rows: 24 | Query complete 00:00:00.198 | CRLF | Ln 4, Col 16

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

11. Create an index on the status column of the flights table. Run EXPLAIN ANALYZE for the query with status = 'Departed' and for the query with status = 'Boarding'



pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM flights
3  WHERE status = 'Departed';
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.19..14.95 rows=5 width=61) (actual time=0.872..0.876 rows=5.00 loops=1)
2	Recheck Cond: ((status)::text = 'Departed'::text)
3	Heap Blocks: exact=1
4	Buffers: shared hit=1 read=1
5	-> Bitmap Index Scan on idx_flights_status (cost=0.00..4.19 rows=5 width=0) (actual time=0.538..0.538 rows=5.00 loo...
6	Index Cond: ((status)::text = 'Departed'::text)
7	Index Searches: 1
8	Buffers: shared read=1

Successfully run. Total query runtime: 157 msec. 12 rows affected.

Total rows: 12 | Query complete 00:00:00.157 | CRLF | Ln 2, Col 10

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions

phonebook
postgres
Login/Group Roles
Tablespaces
PostgreSQL 18

Dashboard × Properties × Statistics × Dependencies × Dependents × Processes × airport_db/postgre... × airport_db/postgres@AZ* ×

airport_db/postgres@AZ

No limit

Query Query History

Scratch Pad ×

```
1  EXPLAIN ANALYZE
2  SELECT * FROM flights
3  WHERE status = 'Boarding';
```

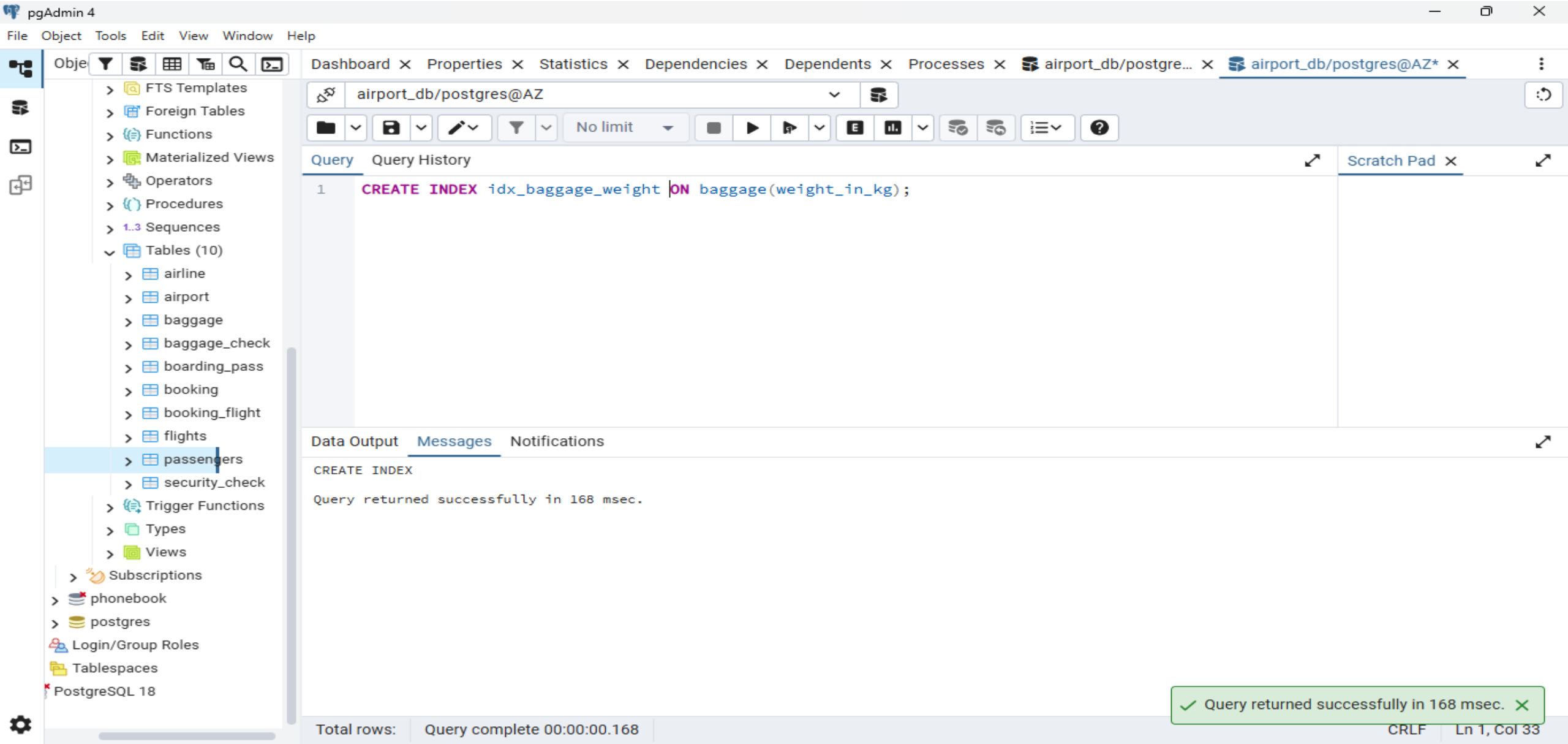
Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on flights (cost=0.00..23.02 rows=399 width=61) (actual time=0.079..0.778 rows=399.00 loop...
2	Filter: ((status)::text = 'Boarding'::text)
3	Rows Removed by Filter: 403
4	Buffers: shared hit=13
5	Planning Time: 0.438 ms
6	Execution Time: 0.884 ms

Total rows: 6 Query complete 00:00:00.291 CRLF Ln 2, Col 10

12. Create an index on the weight_in_kg column of the baggage table. Compare the plans of two range queries: weight_in_kg > 49 (a small part of the table) and weight_in_kg > 5 (almost the whole table)



pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM baggage
3  WHERE weight_in_kg > 49;
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on baggage (cost=0.00..6.70 rows=1 width=22) (actual time=0.068..0.068 rows=0.00 loop...
2	Filter: (weight_in_kg > '49'::numeric)
3	Rows Removed by Filter: 216
4	Buffers: shared hit=4
5	Planning:
6	Buffers: shared hit=2
7	Planning Time: 0.199 ms
8	Execution Time: 0.084 ms

Total rows: 8 | Query complete 00:00:00.177 | CRLF | Ln 2, Col 10

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM baggage
3  WHERE weight_in_kg > 5;
```

Scratch Pad

Data Output | Messages | Notifications

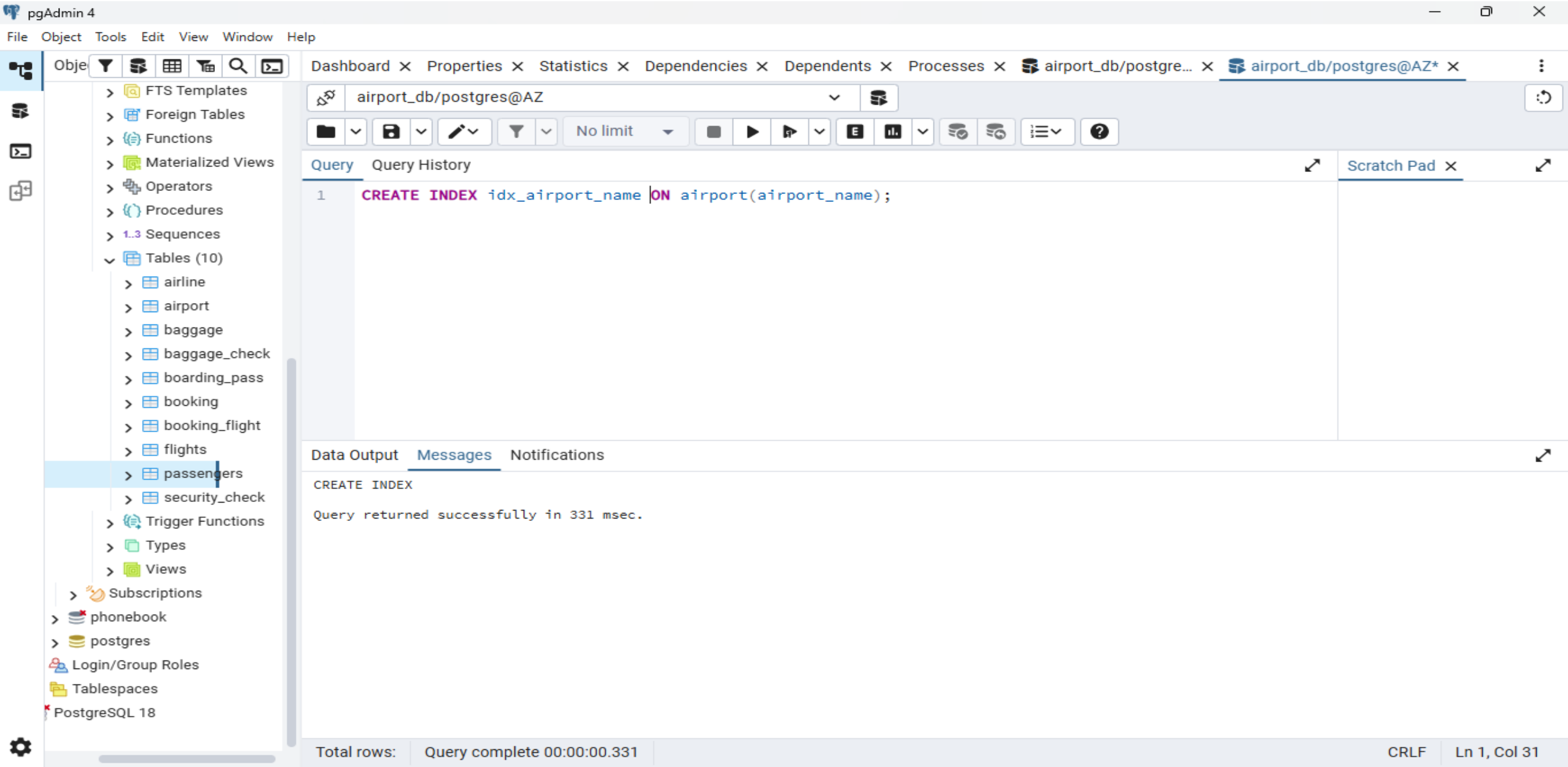
Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on baggage (cost=0.00..6.70 rows=167 width=22) (actual time=0.074..0.246 rows=167.00 loop...
2	Filter: (weight_in_kg > '5'::numeric)
3	Rows Removed by Filter: 49
4	Buffers: shared hit=4
5	Planning Time: 0.379 ms
6	Execution Time: 0.308 ms

Total rows: 6 | Query complete 00:00:00.207 | CRLF | Ln 3, Col 23

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

13. Create an index on the airport_name column of the airport table. Check with EXPLAIN ANALYZE whether it is used by the queries LIKE 'A%' and LIKE '%Airport



pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM airport
3  WHERE airport_name LIKE 'A%';
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.25 rows=1 width=484) (actual time=0.068..0.072 rows=3.00 loop...
2	Filter: ((airport_name)::text ~~ 'A%'::text)
3	Rows Removed by Filter: 17
4	Buffers: shared hit=1
5	Planning:
6	Buffers: shared hit=58 read=7
7	Planning Time: 26.168 ms
8	Execution Time: 0.114 ms

Total rows: 8 | Query complete 00:00:00.174 | CRLF | Ln 2, Col 10

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM airport
3  WHERE airport_name LIKE '%Airport';
```

Scratch Pad

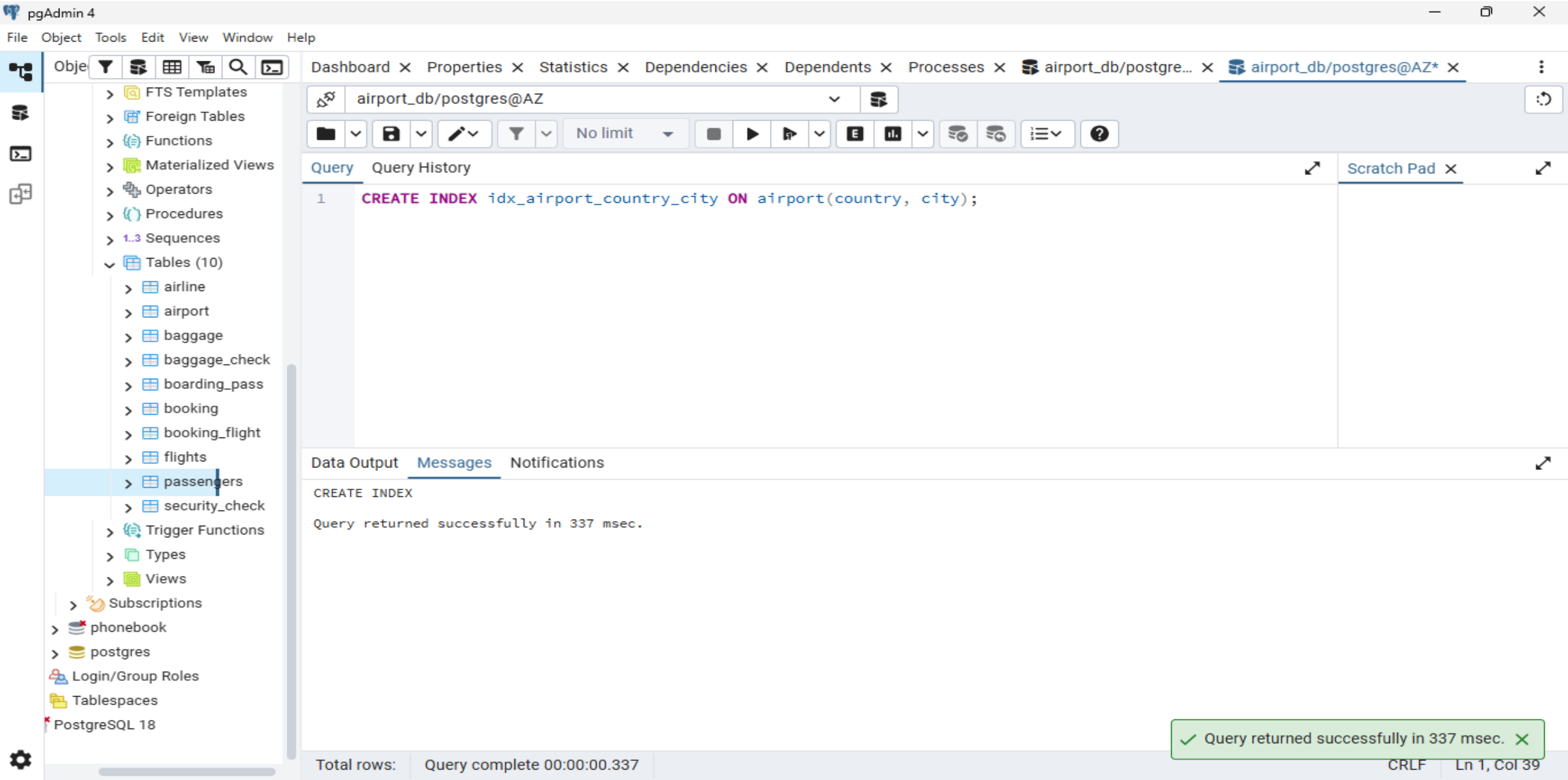
Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.25 rows=1 width=484) (actual time=0.029..0.035 rows=17.00 loop...
2	Filter: ((airport_name)::text ~~ '%Airport'::text)
3	Rows Removed by Filter: 3
4	Buffers: shared hit=1
5	Planning Time: 0.173 ms
6	Execution Time: 0.058 ms

Total rows: 6 | Query complete 00:00:00.210 | CRLF | Ln 2, Col 10

14. Create a composite index on the country and city columns of the airport table. Run three queries: filtering by country only, by city only, and by both columns



pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM airport
3  WHERE country = 'Kazakhstan';
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.25 rows=1 width=484) (actual time=0.090..0.091 rows=0.00 loop...
2	Filter: ((country)::text = 'Kazakhstan'::text)
3	Rows Removed by Filter: 20
4	Buffers: shared hit=1
5	Planning:
6	Buffers: shared hit=20 read=1
7	Planning Time: 8.149 ms
8	Execution Time: 0.178 ms

Total rows: 8 | Query complete 00:00:00.180 | CRLF | Ln 2, Col 10

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Execute script F5

Query Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM airport
3  WHERE city = 'Almaty';
```

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.25 rows=1 width=484) (actual time=0.038..0.038 rows=0.00 loop...
2	Filter: ((city)::text = 'Almaty'::text)
3	Rows Removed by Filter: 20
4	Buffers: shared hit=1
5	Planning Time: 0.189 ms
6	Execution Time: 0.067 ms

Successfully run. Total query runtime: 202 msec. 6 rows affected.

Total rows: 6 Query complete 00:00:00.202 CRLF Ln 2, Col 10

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT *FROM airport
3  WHERE country = 'Kazakhstan' AND city = 'Almaty';
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

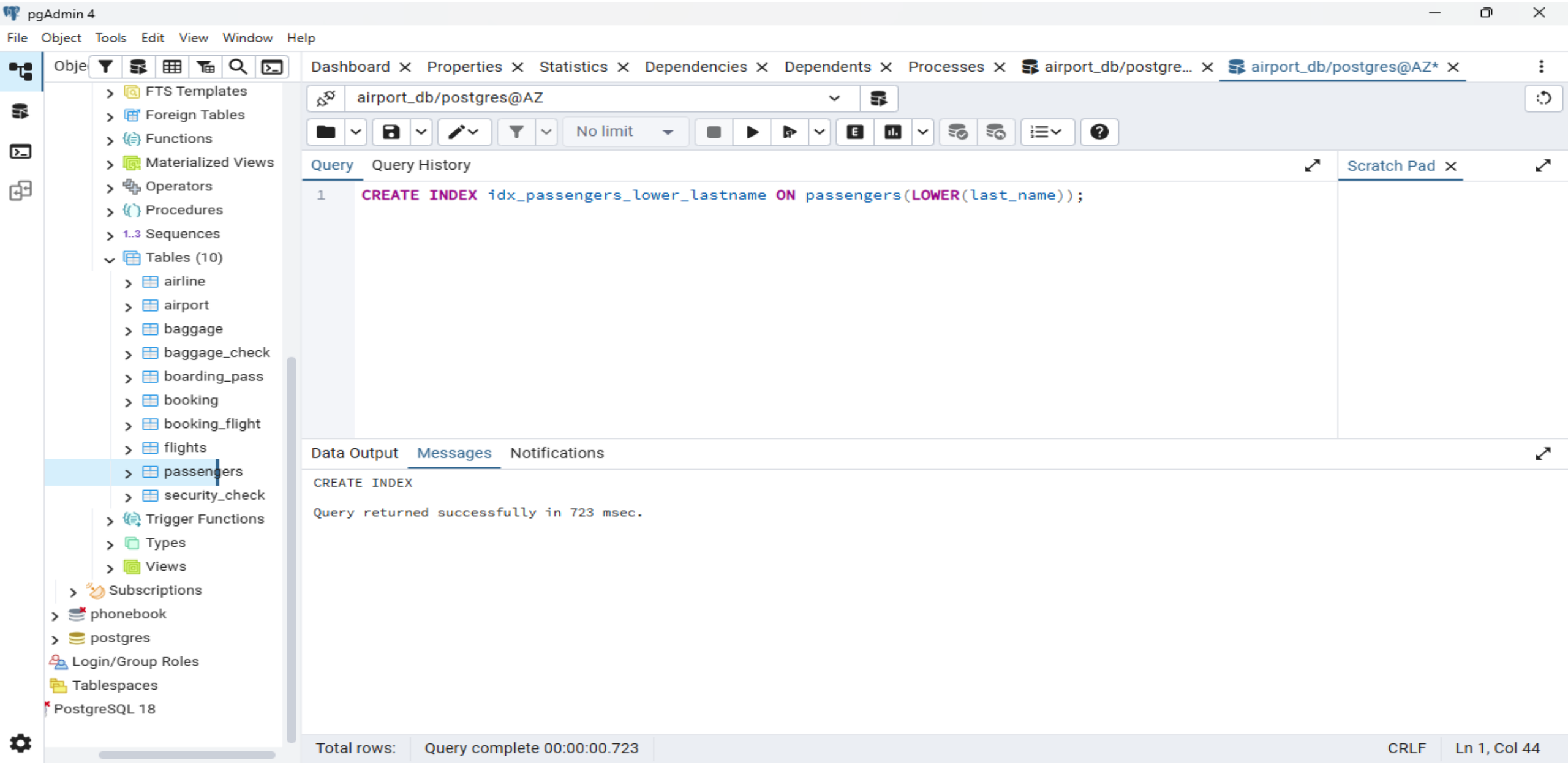
	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.30 rows=1 width=484) (actual time=0.717..0.718 rows=0.00 loop...
2	Filter: (((country)::text = 'Kazakhstan'::text) AND ((city)::text = 'Almaty'::text))
3	Rows Removed by Filter: 20
4	Buffers: shared hit=1
5	Planning Time: 0.186 ms
6	Execution Time: 0.747 ms

Total rows: 6 | Query complete 00:00:00.199 | CRLF | Ln 3, Col 30

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

15. Create an expression (functional) index on the lower-case last_name of the Passengers table (LOWER(last_name)). Compare the plans of a query written as WHERE LOWER(last_name) = 'irnis' and a query written as WHERE last_name = 'Irnis'



pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM passengers
3  WHERE LOWER(last_name) = 'irnis';
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..6.01 rows=1 width=64) (actual time=0.053..0.151 rows=1.00 loop...
2	Filter: (lower((last_name)::text) = 'irnis'::text)
3	Rows Removed by Filter: 200
4	Buffers: shared hit=3
5	Planning:
6	Buffers: shared hit=61 read=4
7	Planning Time: 20.184 ms
8	Execution Time: 0.232 ms

Total rows: 8 | Query complete 00:00:00.178 | CRLF | Ln 2, Col 10

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT * FROM passengers
3  WHERE last_name = 'Irnis';
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..5.51 rows=1 width=64) (actual time=0.075..0.181 rows=1.00 loop...
2	Filter: ((last_name)::text = 'Irnis'::text)
3	Rows Removed by Filter: 200
4	Buffers: shared hit=3
5	Planning Time: 0.335 ms
6	Execution Time: 0.233 ms

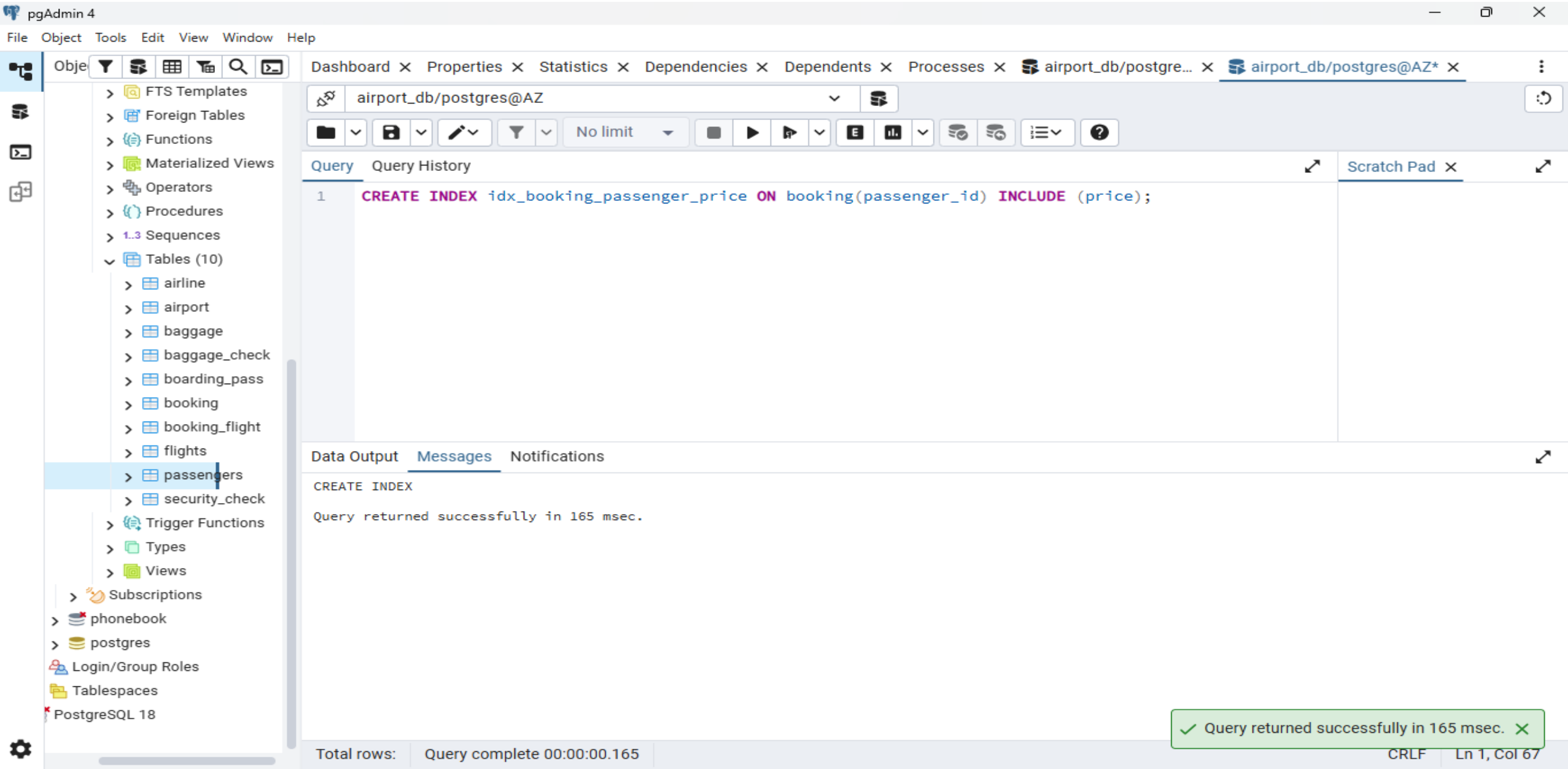
Successfully run. Total query runtime: 1 secs 991 msec. 6 rows affected.

Total rows: 6 | Query complete 00:00:01.991 | CRLF | Ln 2, Col 10

Obje

- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
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- Tablespaces
- PostgreSQL 18

16. Create a covering index on the booking table for the column passenger_id and price. Write a query that returns only price for a given passenger_id and check with EXPLAIN ANALYZE whether an index-only scan (or "Using index") is used.



pgAdmin 4

File Object Tools Edit View Window Help

Obje

- FTS Templates
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- Tables (10)
  - airline
  - airport
  - baggage
  - baggage_check
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  - booking_flight
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  - passengers
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- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook
- postgres
- Login/Group Roles
- Tablespaces
- PostgreSQL 18

Dashboard | Properties | Statistics | Dependencies | Dependents | Processes | airport_db/postgre... | airport_db/postgres@AZ*

airport_db/postgres@AZ

No limit

Query | Query History

```
1  EXPLAIN ANALYZE
2  SELECT price FROM booking
3  WHERE passenger_id = 100;
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
5	-> Bitmap Index Scan on idx_booking_passenger (cost=0.00..4.17 rows=3 width=0) (actual time=0.741..0.741 rows=3.00 loo...
6	Index Cond: (passenger_id = 100)
7	Index Searches: 1
8	Buffers: shared hit=1
9	Planning:
10	Buffers: shared hit=20 read=1 dirtied=3
11	Planning Time: 4.997 ms
12	Execution Time: 1.445 ms

Total rows: 12 | Query complete 00:00:00.151 | CRLF | Ln 2, Col 14